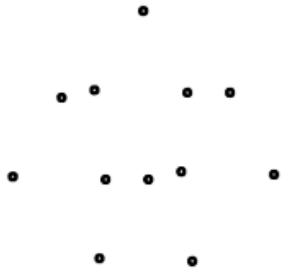
Event Control

New Methods for Cyber Security
The Problem



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View as a slideshow for animation support.

Event Control

Knowledge to Action

- Methods to increase awareness of system events for better system control
- Cyber Security Application

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The Problem

- The increased complexity and scale of modern networked systems leads to the loss of control, despite the most heroic management efforts.
- There are the typical problems with staffing, budget, executive buy-in, integration and implementation issues, etc…
- But there is a more fundamental problem at work that makes these problems more intractable.
- If you think of networks as physical systems, behaving like other physical phenomena, we can learn something from well-established physical laws. One of the most dependable and inviolable these is the 2nd Law of Thermodynamics:

https://en.wikipedia.org/wiki/Second_law_of_thermodynamics ▾

The **second law of thermodynamics** states that the total entropy of an isolated system can never decrease over time.

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And What This Guy Says!

The law that entropy always increases, holds, I think, the supreme position among the laws of Nature. If someone points out to you that your pet theory of the universe is in disagreement with Maxwell's equations — then so much the worse for Maxwell's equations. If it is found to be contradicted by observation — well, these experimentalists do bungle things sometimes. But if your theory is found to be against the second law of thermodynamics I can give you no hope; there is nothing for it but to collapse in deepest humiliation.

Sir Arthur Stanley Eddington

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What Does This Have To Do with Cyber Security?

- An “isolated system” is a system that has no inputs i.e. is not managed.
- “Entropy” is the degree of disorder in the system aka “my network is FUBAR… I mean, is displaying high levels of entropy”
- In order to decrease entropy, increasing levels of management must be applied to a system.
- However the “management” is also a system that must have continuous inputs to avoid increased entropy… and so on ad infinitum.
- Management “systems” also increase the overall tendency towards entropy… those have to be managed too.
- As Eddington and The Notorious B.I.G said, there’s just no way around it: “Mo’ money, Mo’ problems”

+

= Seems pretty legit

ToC

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But Wait! Problem #2

- We are heading full speed into the unfamiliar world of Big Data.
- We are not yet equipped to make sense of it intuitively.
- We tend towards a logical fallacy called The Base Rate Fallacy. Our minds tend to ignore generalities when confronted with specifics. We project the specifics we know onto the unknown, which can be very misleading
- Big Data is… big. Vast. Unknown. We look at small pieces and can't help but extrapolate them into some kind of big picture for ourselves.
- Bayes Theorem of Conditional Probability demonstrates this mathematically (more on this later).

$$P(A|B) = \frac{P(B|A)P(A)}{P(B)}$$

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So…

- A complex system tends toward disorder
- The addition of mitigating controls into a system tends to increase the overall complexity of the system and therefor often just compounds the problem
- We don’t have a good intuitive sense of the character of the data we are now trying to understand.

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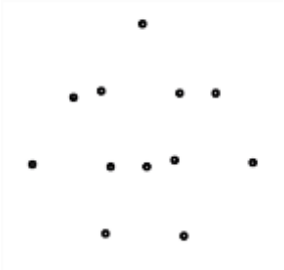
Thank You



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